

Taylor D. Sparks

materials chemistry, at career year 6

1,091

people in the cohort

6

career year both sides are
measured at

2008 to 2018

when the cohort started

6

OpenAlex topics defining
the subfield

Taylor D. Sparks started a tenure-line appointment at University of Utah in 2013. Everyone in the cohort is estimated to have started one in materials chemistry between 2008 and 2018, so both sides can be read at the same point on the clock. Taylor D. Sparks is not in the cohort.

These numbers describe what a group of people did. They are not a standard, nobody in the cohort agreed to be measured, and no part of this says what any one career should look like.

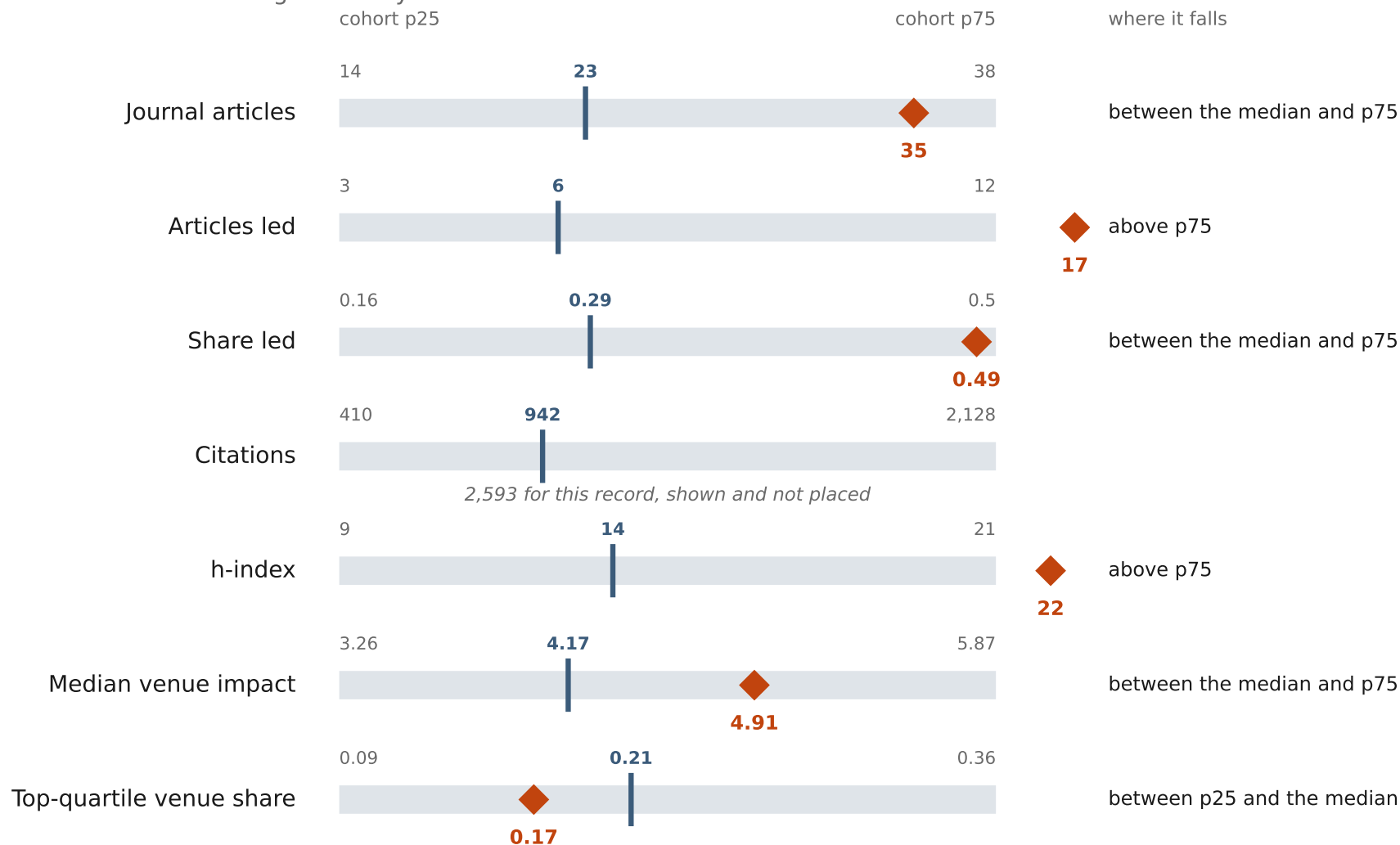
WHAT IS IN HERE

- | | | |
|----------|------------------------------|---|
| 1 | Where this record sits | each measure against the cohort's middle half |
| 2 | Year by year | how the cohort's record grew across the clock |
| 3 | The numbers | quartiles, with confidence intervals |
| 4 | Who is in the cohort | every filter, and how many people it left |
| 5 | Where the subfield publishes | which journals top quartile covers |
| 6 | The chaperone effect | who leads the papers that reach the best venues |
| 7 | What this cannot see | the limits, in plain words |

1. AGAINST THE COHORT

Where this record sits at year 6

The shaded band is the middle half of the cohort, p25 to p75, with the median marked. The diamond is Taylor D. Sparks. Both sides are counted through career year 6.

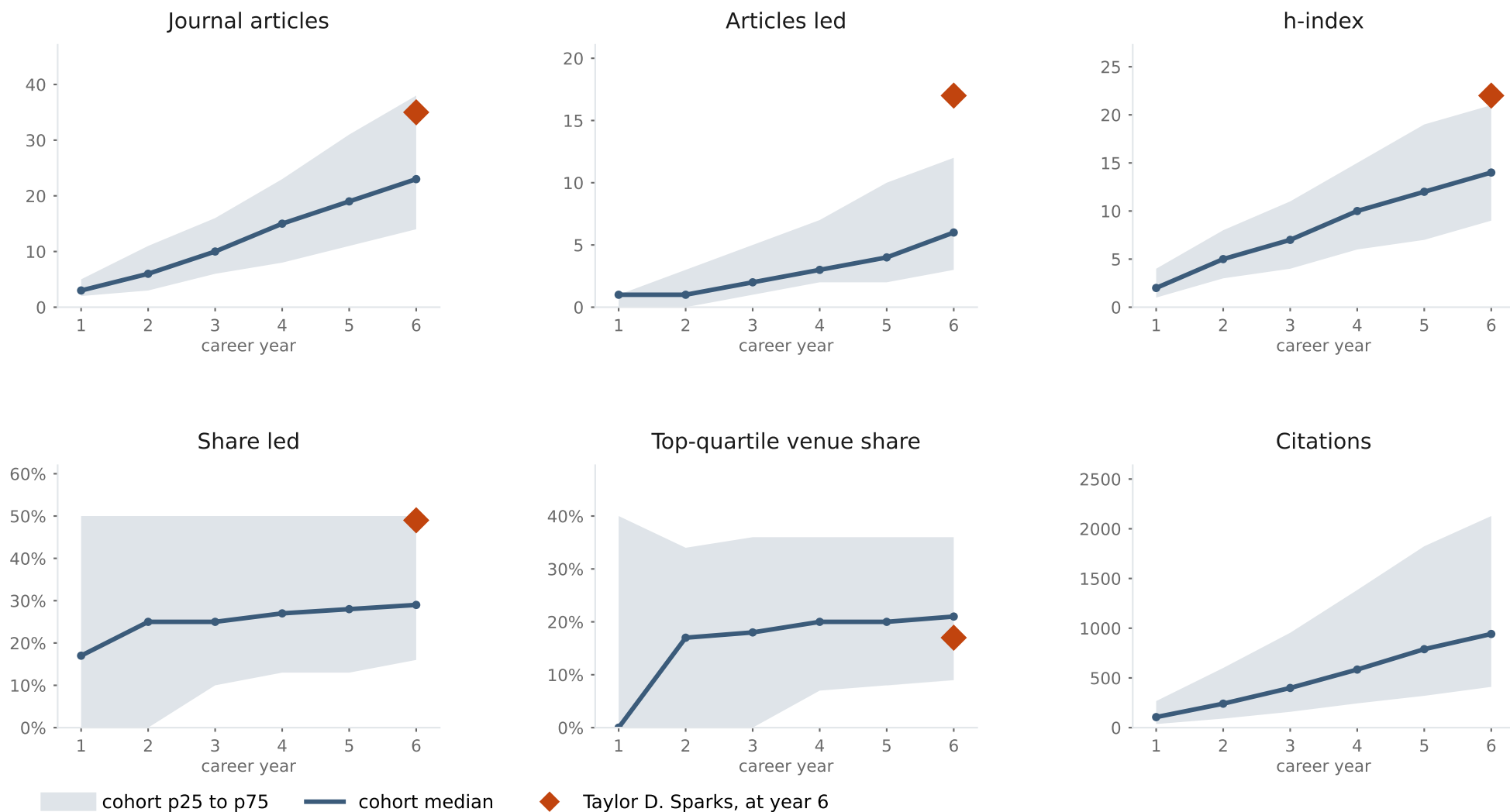


A value past p25 or p75 is drawn just outside the band rather than to scale, so one long reach does not squash every other row. Citations carry a count and no position: the cohort's papers in this window are eight to eighteen years old and this record's are at most 6 years old, so setting the two counts side by side would measure the calendar rather than the work.

2. ACROSS THE CLOCK

The same cohort, year by year

Each panel is one measure counted from the start of the appointment through that career year. The band is the cohort's middle half and the line is its median. Taylor is marked at year 6, the year the two sides are compared at.



Counts are cumulative, so a flat stretch is a year that added little and a step is a year that added a lot. Citations are counted as they stand today for the papers published by that career year, which is why the early years look larger here than they did at the time.

3. THE COHORT AT YEAR 6

The numbers those bands come from

What the cohort of 1,091 people had published by the end of career year 6, as quartiles across people.

Measured through year 6	Lower quarter (p25)	Median	Upper quarter (p75)
Journal articles	14 (13 to 14)	23 (22 to 25)	38 (36 to 40)
Led articles (last or corresponding)	3 (3 to 3)	6 (5 to 6)	12 (11 to 13)
Share of articles led	0.16 (0.14 to 0.17)	0.29 (0.27 to 0.31)	0.5 (0.46 to 0.5)
Citations to those articles	410 (362 to 448)	942 (853 to 1,036)	2,128 (1,940 to 2,394)
h-index over those articles	9 (8 to 9)	14 (13 to 15)	21 (20 to 22)
Median venue impact	3.26 (3.13 to 3.44)	4.17 (3.97 to 4.33)	5.87 (5.64 to 5.93)
Share in top-quartile venues	0.09 (0.08 to 0.11)	0.21 (0.2 to 0.23)	0.36 (0.33 to 0.38)

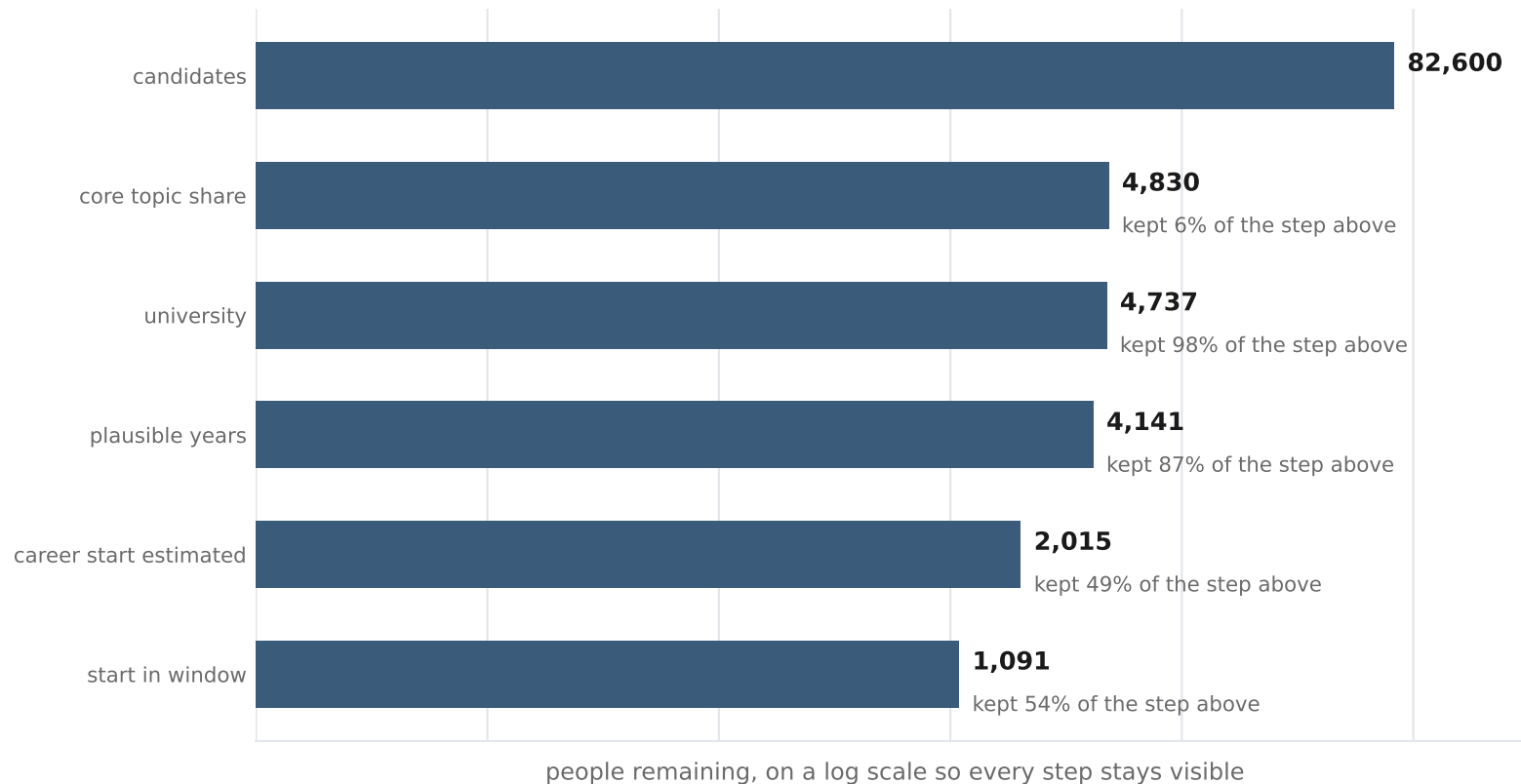
Read a row like this: a quarter of the cohort was below the first number, half was below the middle one, and a quarter was above the last. Parentheses are 95% confidence intervals from a cluster bootstrap that resamples people rather than papers, because one person's papers are not independent of each other. A wide interval means the cohort is small or spread out, and the number should be leaned on lightly.

A top-quartile venue is top quartile inside this cohort's own venue list, not against a global journal ranking, because citation rates differ enormously between subfields. The journals it covers are named later in this report.

4. HOW IT WAS BUILT

Who ended up in the cohort

This is the page to read first when a cohort looks wrong. A step that removes almost everybody, or almost nobody, is usually the one to question.



candidates: topics T11948|T10440|T10657|T10018|T10311|T10059, at least 10 works, an affiliation in ...

core topic share: share of work in the subfield at least 0.4

university: an affiliation of type education

plausible years: byline years could contain a start between 2008 and 2018

career start estimated: a confident first independent start (at least 2 led papers at one institution, with ...

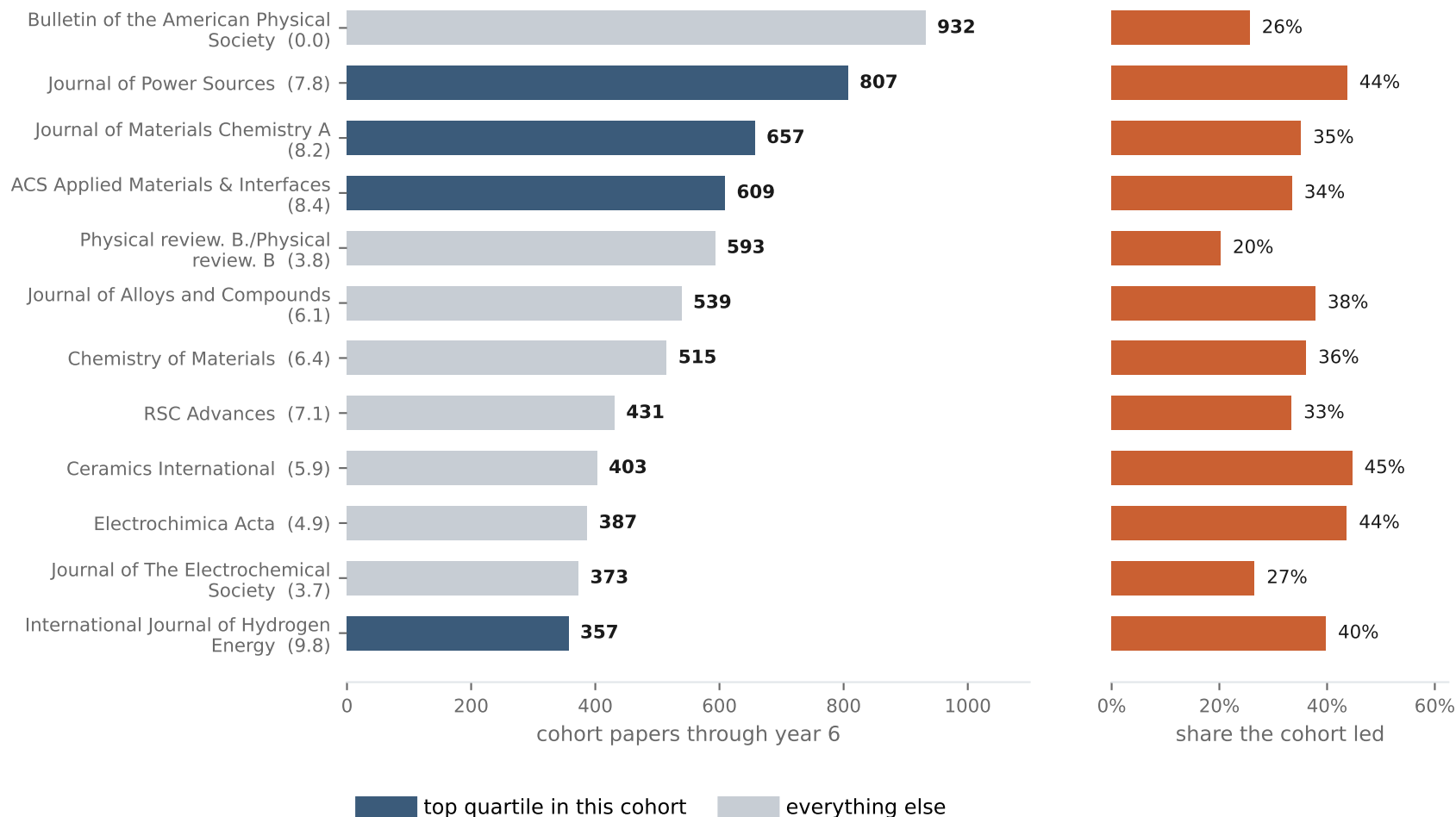
start in window: estimated start between 2008 and 2018

The full rule for each step is in results/funnel.csv and in docs/methods.md. Where a person could not be placed with confidence they were dropped rather than guessed at, so the cohort is smaller and cleaner than the subfield it came from, and the people it leaves out are not a random sample of that subfield.

5. THE VENUES

Where this subfield publishes

The journals the cohort used most, so that a top-quartile venue can be checked against titles rather than taken on trust. The panel on the right is how much of that work the cohort led.



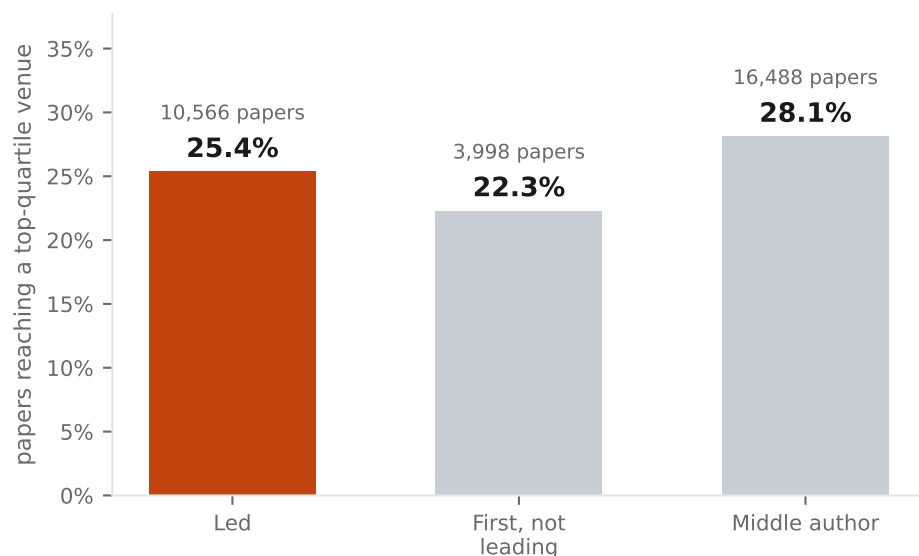
The number in brackets after a journal is its 2-year mean citedness from OpenAlex, a property of the journal and not of any paper in it. Read this list before trusting the counts elsewhere in the report. Some conference abstract series carry an ISSN and are typed as journals by OpenAlex, so they cannot be told apart from a journal in the data and are counted as articles here. A journal near the top of this list with an impact near zero is usually one of them.

6. THE CHAPERONE EFFECT

Who leads the papers that reach the best venues

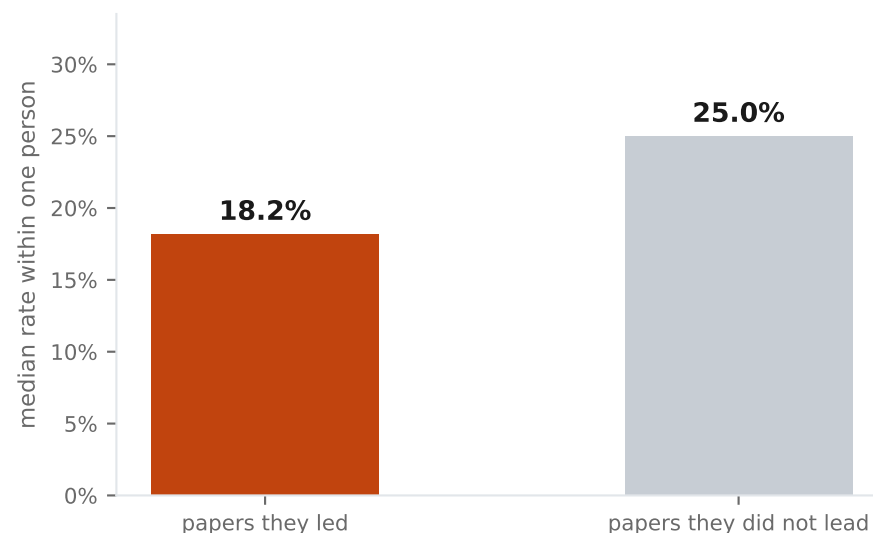
When a paper reached a top-quartile venue, was this person leading it or co-authoring somebody else's? Sekara et al. called that second route the chaperone effect. Led means last author, or flagged as the corresponding one.

Across every paper the cohort wrote



Every paper counts once, so this reading is dominated by whoever wrote the most of them.

The same 770 people, each their own comparison



Everyone here is their own control, so field, institution, career stage and how prolific someone is all cancel out. It drops anyone without enough papers in both roles.

Pooled across every paper, the cohort's work reached a top-quartile venue 2.8% more often when its members were not leading it (95% confidence interval +0.6% to +5.0%). The interval comes from a cluster bootstrap that resamples people, so it says how far this would move with a different draw of people from the same subfield. Where the two panels disagree, that disagreement is the finding.

After Sekara et al., "The chaperone effect in scientific publishing", PNAS 2018 (doi 10.1073/pnas.1800471115), which followed authors through time. This is a cross-sectional approximation of that design and not a replication: the direction of a difference is informative, its size should not be read against their figures. Where a field does not order authors by contribution, none of this applies.

LIMITS

What this cannot see

Six things to hold in mind before this document is used for anything.

A publication record is not a person.

Teaching, mentoring, service, funding, software, datasets, patents and public scholarship do not appear in OpenAlex, and they are a large part of the job.

Citations are shown and never placed.

The cohort's papers in this window are years older than this record's, and citations accumulate with time, so comparing the two counts would measure the calendar.

Career start is inferred, not looked up.

It comes from publication patterns rather than from HR records. Conversions from lecturer to tenure line, clinical appointments, parental leave and delayed starts are invisible to it, and people who never changed institution are left out because their trainee years cannot be told apart from their independent ones.

The people who are missing are not a random sample.

OpenAlex splits some people across profiles and merges others with namesakes. The cohort keeps only the people it could identify confidently, which tilts it toward distinctive names.

Journal impact says nothing about a paper.

It is a property of the journal. Venue quartiles are computed inside this cohort's own venue list because citation cultures differ enormously between subfields.

This is a description, not an instruction.

These are ranges of what a group of people did. A department that reads the median as something everyone must reach has misread this document.